

The Layers of Cake-Cutting

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1 Introduction

A birthday is a day of celebration. Whether you're enjoying one with friends, family, or a mix of both, birthdays provide a semblance of unity entwined with festivity... until the cake is pulled out.

It's a tale as old as time. Everybody sights out their own desired slice of cake. A couple partygoers aim for the strawberry at the center, while another few desire the delicately swirled puffs of meringue frosting; nobody wants to walk away with a plain slice. Unfortunately, there often isn't enough cake to get picky. Though quarreling over slice preference seems inevitable with every birthday celebration, mathematicians have been steadily working towards an algorithm that would finally facilitate a utopian party, where everyone leaves with what they want.

The aptly-named cake-cutting problem defines a scenario where a heterogenous, finite object is split among a number of people, each valuing their own pieces of the object. Because of the scenario's applicability to the real world, from territorial disputes to birthday parties, the cake-cutting problem has been cracked at by mathematicians for generations. With modern algorithms to solve the problem having been largely derived through software programs, the efficiency of these algorithms has been steadily becoming more refined.

The cake-cutting problem hinges on the game theory concept of "envy-freeness." That is, in a fair division situation, n players intend to obtain a piece of an object (denoted as C) such that each participant is satisfied that their share is as sufficient as any other player's share. Given the subjectivity of how much a player values an element of the object, desirability is expressed with the value function, V_x , for player x . If player x favors their own share (C_x) over player y 's share (C_y), then x is envy-free:

$$V_x(C_x) \geq V_x(C_y)$$

Conversely, if player x favors player y 's share over their own, x is considered envious of y :

$$V_x(C_x) < V_x(C_y)$$

In an envy-free system, no player is envious of each other.

The modern mathematical concept of an envy-free cake-splitting division was defined in 1958 by polymath George Glamow and Marvin Stern. Since then, the problem's prevalence in resource division launched it into prominence as an optimizable algorithm. By the 1960s, algorithms for fair division among 2 and 3 players had been established.

For $n = 2$, the simplistic "divide and choose" process is employed; player 1 begins by dividing the cake into two pieces that they value equally. Then, player 2 selects the piece that they value

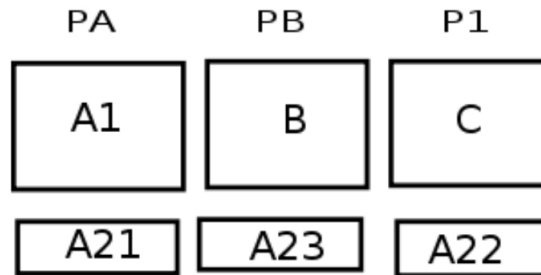
more (C_2), leaving the other piece (C_1) for player 1. Therefore:

$$V_1(C_1) = V_1(C_2) \text{ and } V_2(C_2) \geq V_2(C_1)$$

This satisfies the condition of envy-freeness, as both players value their piece as much or more than the other piece.

The Selfridge-Conway procedure, discovered in 1960 by mathematician John Selfridge, allows for envy-free division among 3 players. Notably, this algorithm assumes the cake can be split into “general pieces”, where a sum of pieces can constitute a player’s share. This contrasts from a “connected pieces” scenario, where n pieces must be available for n players, and each player must receive one contiguous piece (therefore meaning that $n - 1$ cuts can be performed). The Selfridge-Conway procedure has an upper bound of 5 cuts, and is as follows:

1. Akin to the “divide and choose” algorithm, player 1 begins by dividing the cake into three slices that they perceive as equal.
2. Player 2 trims off part of their most valued slice (C_A) so that player 2 values it equally as much as the next most valuable slice. The trimmings are put aside for later. Therefore, C_A is now divided into a trimmed slice (C_{A1}) and trimmings (C_{A2}).
3. Player 3 then chooses any slice.
4. If player 3 did not choose C_{A1} , player 2 must select it. Otherwise, player 2 takes any slice.
5. Player 1 takes the remaining slice, leaving only the trimmings. Now, the trimmings must be divided. Whichever player took the slice C_{A1} is denoted as player A , and the other player as player B .
6. Player B divides the trimmings into three equally-valued slices.
7. Player A chooses a slice of the trimmings (C_{A21}).
8. Player 1 chooses a slice of the trimmings (C_{A22}).
9. Player B chooses the last trimming slice (C_{A23}).



Player A has C_{A1} and C_{A21} . Player B has C_B and C_{A23} . Player 1 has C_C and C_{A22} . Player A is satisfied because:

- $V_a(C_{A1}) > V_A(C_B)$
- $V_a(C_{A1}) > V_A(C_C)$

- $V_a(C_{A21}) > V_A(C_{A23})$
- $V_a(C_{A21}) > V_A(C_{A22})$
- Therefore, $V_A(C_{A1} + C_{A21}) \geq V_A(C_B + C_{A23}), C_A(C_C + C_{A22})$

Player B is satisfied because:

- $V_B(C_B) \geq V_B(C_{A1})$
- $V_B(C_B) \geq V_B(C_C)$
- $V_B(C_{A23}) = V_B(C_{A22}) = V_B(C_{A21})$
- Therefore, $V_B(C_B + C_{A23}) \geq V_B(C_{A1} + C_{A21}), V_B(C_C + C_{A22})$.

Player C is satisfied because:

- $V_C(C_C) \geq V_C(C_{A1} + C_{A21})$
- $V_C(C_C) = V_C(C_B)$
- $V_C(C_{A22}) \geq V_C(C_{A23})$
- Therefore, $V_C(C_C + C_{A22}) \geq V_C(C_B + C_{A23}), V_C(C_{A1} + C_{A21})$.

This scenario fulfills envy-freeness; ultimately, each player is satisfied with their combination of piece and trimming.

That was a doozy.

Since the 1960s, the cake-splitting problem has continued to see development. In 1995, mathematician Alan Taylor and political scientist Steven Brams deduced a procedure that promised envy-freeness. However, the unbounded nature of the algorithm meant that for complicated scenarios, runtime could exceed billions of steps based on the number of players and the specificity of their preferences.

Further progress in 2016 by computer scientists Simon Mackenzie and Haris Aziz reduced the upper bound to only depend on the number of players. Using the trim-dividing premise of the Selfridge-Conway procedure in conjunction with novel concepts of player prioritization, Mackenzie and Aziz were able to decrease the number of players requiring cake with each subsequent step. Though the algorithm's hefty exponential runtime may render it unfit for your next birthday dispute, the cake-splitting problem's applicable zaniness has underscored one of the most fascinating slices of modern game theory.

References

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